

3.9 Organic Synthesis

Question Paper

Course	CIEA Level Chemistry
Section	3. Organic Chemistry
Topic	3.9 Organic Synthesis
Difficulty	Medium

Time allowed: 70

Score: /54

Percentage: /100

Question 1a

But-2-ene can be converted into butan-2-ol by two different synthetic routes. One route can be carried out as a one-step process whilst the other requires two steps.

State reagents and provide an equation, using structural formulae, for the one-step process.

[3 marks]

Question 1b

Draw a reaction scheme, including reagents and conditions for the **two** step process.

[3 marks]

Question 1c

But-1-ene can be used to form butan-2-ol by both the one-step and two-step reaction schemes. Explain why using but-1-ene gives a lower yield of butan-2-ol.

[2 marks]

Question 1d

Give the displayed formula of the organic compound(s) formed when but-2-ene is

- reacted with cold, dilute MnO_4^- ions,
- reacted with hot, concentrated MnO_4^- ions

[2 marks]

Question 2a

A series of reactions based on butan-1-ol is shown in Fig. 2.1.

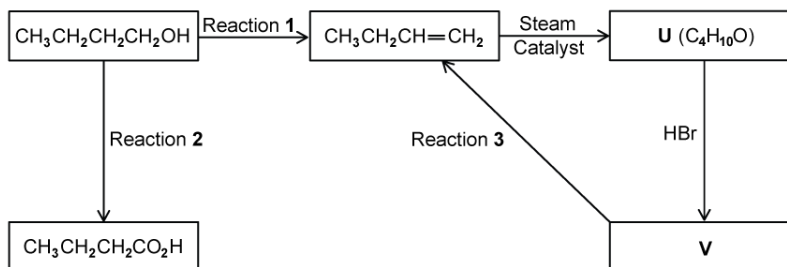


Fig. 2.1

Suggest a suitable reagent and conditions for reaction 1.

[2 marks]

Question 2b

i)

Write an equation for reaction 2, using [O] to represent the oxidising agent.

[1]

ii)

Suggest a suitable reagent and conditions for reaction 2.

[2]

[3 marks]

Question 2cGive the structural formulae of **U** and **V**.**U****V**

[2 marks]

Question 2dSuggest a suitable reagent and conditions for reaction **3**.

[2 marks]

Some reactions of compound A are shown in Fig. 3.1.

Fig. 3.1

[3 marks]

State the colour change when compound **D** is formed.

[1 mark]

Question 3c

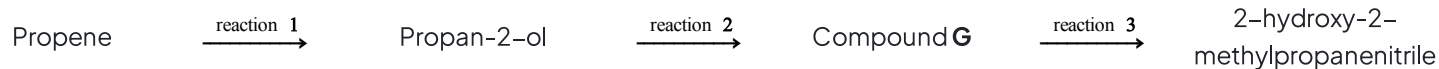
Compound **A** reacts with iodine and sodium hydroxide ions to give two products. Draw the skeletal structures of these **two** products.

[2 marks]**Question 3d**

State the name of compound **A**.

[1 mark]**Question 4a**

The reaction scheme shows three steps to synthesise 2-hydroxy-2-methylpropanenitrile.



Name the type of mechanism that occurs in

- reaction 1
- reaction 3

[2 marks]**Question 4b**

State the reagents and conditions required for

- reaction 1
- reaction 2

[4 marks]

Question 4c

Draw the mechanism that occurs in reaction 3

[4 marks]

Question 4d

Explain why 2-hydroxy-2-methylpropanenitrile does not exhibit optical isomerism.

[2 marks]

Question 5a

The following reaction pathway in Fig. 5.1 is used to produce compounds **A** and **B**, which when reacted together, form a branched ester molecule, compound **C**.

Suggest suitable reagents and conditions for the synthesis of compound **A** via step 1 and give the name for this type of reaction.

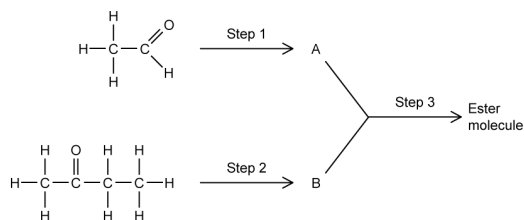


Fig. 5.1

[3 marks]

Question 5b

In order for the ester to be produced, the ketone in part (a) must be converted to another compound, **B**.

- i)
Name and draw the structure of the molecule that is produced from step 2.

[2]

- ii)
Give the name of the type of reaction that is involved in step 2 and suggest suitable reagents and conditions for the process.

[2]

[4 marks]

Question 5c

Outline how ethanol can be synthesised from ethane in **two** steps. State the reaction conditions and reagents and name the type of reaction taking place.

[6 marks]

Question 5d

The four step synthesis to form propan-1-ol from a ketone is outlined below in Fig. 5.2.

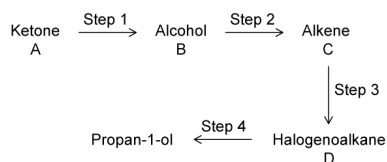


Fig. 5.2

i)

Give the names of four possible substances **A** to **D**

[1]

ii)

Give the reagents and conditions for step 4.

[2]

[3 marks]

Model Answers: Medium

1a

a) The reagents and equation for the one-step process converting but-2-ene into butan-2-ol are:

Reagents:

- Steam / water vapour / H_2O (g); [1 mark]
- Phosphoric acid / H_3PO_4 catalyst

OR

Sulfuric acid / H_2SO_4 catalyst; [1 mark]

Equation:

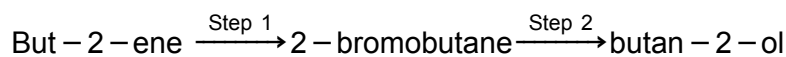
- $\text{CH}_3\text{CHCHCH}_3 + \text{H}_2\text{O} \rightarrow \text{CH}_3\text{CHOHCH}_2\text{CH}_3$; [1 mark]

[Total: 3 marks]

- The reaction converting an alkene directly into an alcohol is hydration requiring steam and a strong acid catalyst such as phosphoric or sulfuric acid
- This can also be described as electrophilic addition

1b

b) The reaction scheme, including reagents and conditions for the **two** step process is:



Intermediate compound:

- 2-bromobutane; [1 mark]

Step 1 reagent:

- Hydrogen bromide; [1 mark]

Step 2 reagent:

- Aqueous sodium hydroxide / NaOH (aq).
- OR**
- Aqueous potassium hydroxide / KOH (aq); [1 mark]

[Total: 3 marks]

- This reaction uses HBr as the electrophile
- It is an electrophilic addition reaction involving heterolytic bond fission

1c

c) Using but-1-ene gives a lower yield of butan-2-ol because:

- (When the alkene double bond breaks open in but-1-ene) it can form a primary and a secondary carbocation intermediate; [1 mark]
- (Resulting in the) formation of butan-1-ol and butan-2-ol; [1 mark]

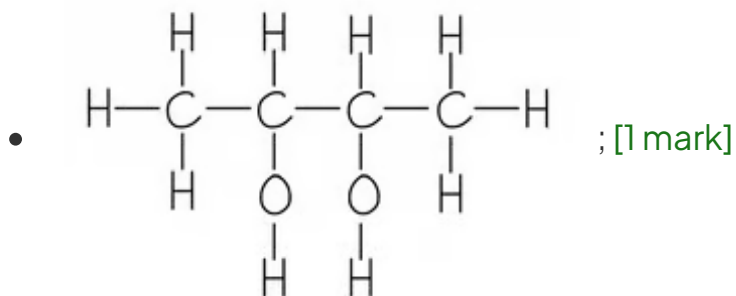
[Total: 2 marks]

- But-2-ene is a symmetrical alkene and will only form butan-2-ol
- But-1-ene is not a symmetrical alkene and can form two carbocation intermediates
 - This results in the formation of a major and a minor product
 - You should be able to apply and explain Markownikoff's rule in writing as well as by drawing mechanisms

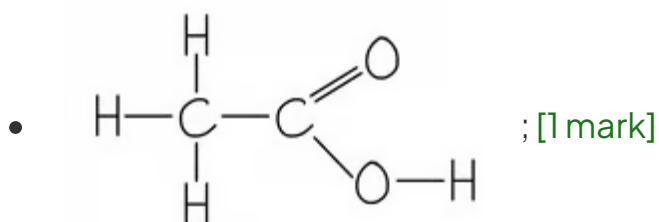
1d

d) The displayed formula for both reactions are:

Reaction with cold, dilute MnO_4^- ions:

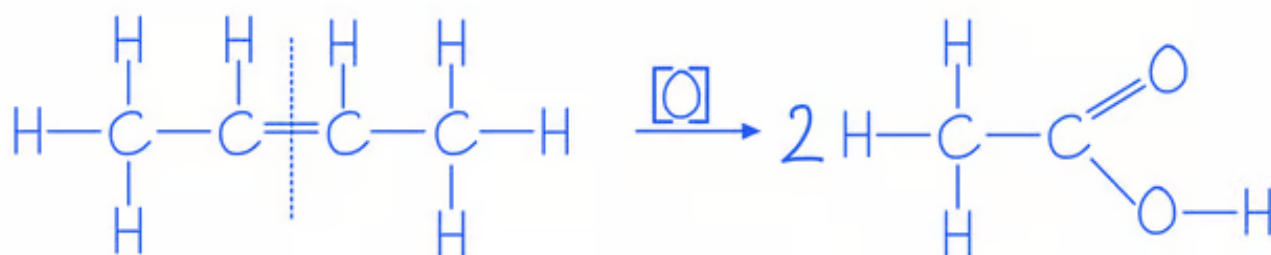


Reaction with hot, concentrated MnO_4^- ions:



[Total: 2 marks]

- You should know the reactions of alkenes with both cold dilute KMnO_4 and hot concentrated KMnO_4
- The reaction of hot concentrated KMnO_4 with an alkene results in the breaking of the double bond
- As there is only one CH_3 group either side of the double bond, two molecules of ethanoic acid are formed
 - Therefore there is only one type of compound formed from this reaction



- **Careful:** You are asked for the displayed formulas, so the OH group should be drawn as O-H

2a

a) A suitable reagent and conditions for reaction 1 is:

- Reagents = conc H_2SO_4 / conc H_3PO_4 ; [1 mark]
- Conditions = heat

OR

Pass vapour over hot Al_2O_3 ; [1 mark]

[Total: 2 marks]

- Reaction 1 is the dehydration of an alcohol which forms an alkene

- Dehydration is a reaction in which a water molecule is removed from a larger molecule
- A dehydration reaction is a type of elimination reaction
- Look for the changes in the compound to help you identify suitable reagents and conditions

2b

b)

i) An equation for reaction **2**, using [O] to represent the oxidising agent

- $\text{C}_4\text{H}_9\text{OH} + 2[\text{O}] \rightarrow \text{C}_3\text{H}_7\text{CO}_2\text{H} + \text{H}_2\text{O}$; [1 mark]

ii) A suitable reagent and conditions for reaction **2**.

- Reagent = sodium dichromate / $\text{Na}_2\text{Cr}_2\text{O}_7$

OR

Potassium dichromate / $\text{K}_2\text{Cr}_2\text{O}_7$; [1 mark]

- Conditions = H^+ / acidified

AND

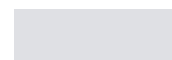
(Heat under) reflux; [1 mark]

[Total: 3 marks]

- Reaction 2 is the oxidation of butan-1-ol to form butanoic acid
- Primary alcohols oxidise to form aldehydes and then carboxylic acids
- This is the full oxidation, as the reaction scheme does not show the aldehyde has been produced
- If the product was the aldehyde then distillation apparatus would be used, not reflux

2c

c) The structures of U and V are:



Potassium dichromate / $K_2Cr_2O_7$; [1 mark]

- Conditions = H^+ / acidified

AND

(Heat under) reflux; [1 mark]

[Total: 3 marks]

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2c

c) The structures of U and V are:

- **U** = $CH_3CH_2CH(OH)CH_3$

OR

U = $CH_3CH_2CH_2CH_2OH$; [1 mark]

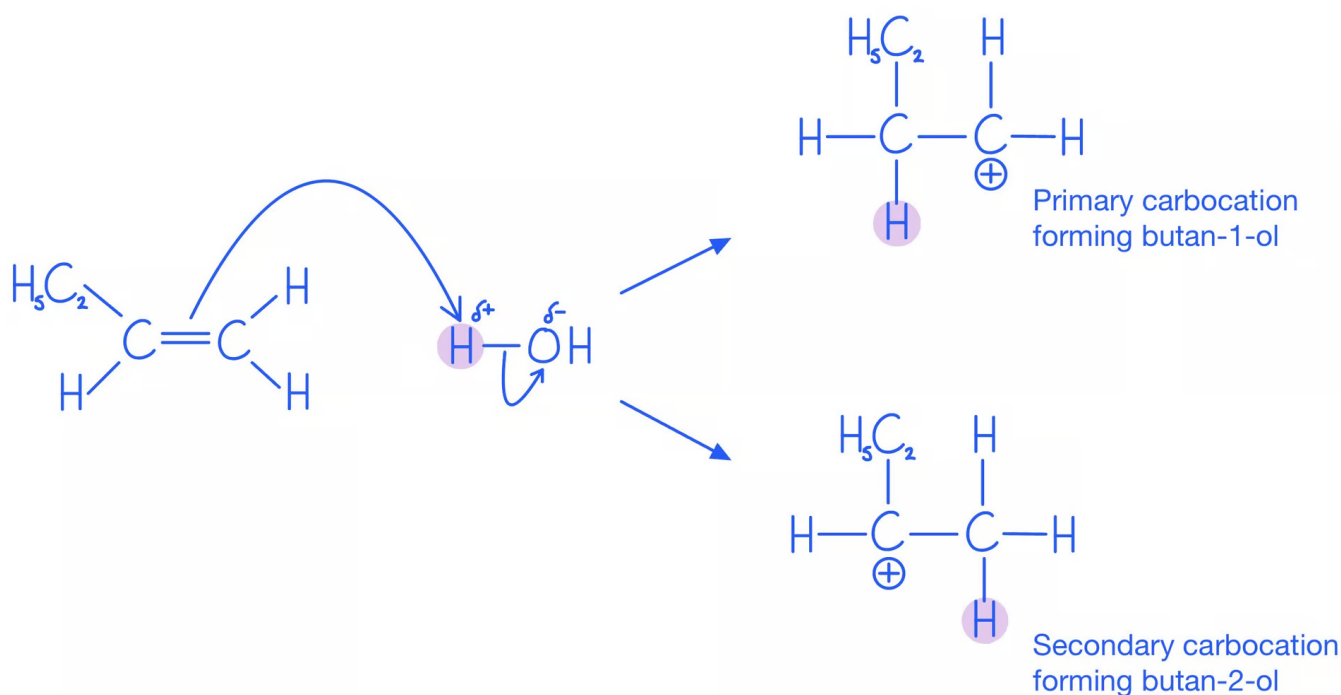
- **V** = $CH_3CH_2CHBrCH_3$

OR

V = $CH_3CH_2CH_2CH_2Br$; [1 mark]

[Total: 2 marks]

- Reacting but-1-ene with steam will form an alcohol
 - A secondary alcohol will be the major product, and the primary alcohol will be the minor product
 - This is due to the difference in stability of the primary and secondary carbocation
- The first step of this addition mechanism is shown below:



- It is not specified if a minor or major product is formed so it doesn't matter which one you choose
- The primary alcohol will then undergo a nucleophilic substitution reaction with HBr and form 1-bromobutane
- The secondary alcohol will react in the same way but will form 2-bromobutane

2d

d) Suitable reagent and conditions for reaction **3** are:

- Reagent = KOH / NaOH; [1 mark]
 - Conditions = ethanol / alcohol
- AND**
- Heat / reflux; [1 mark]

[Total: 2 marks]

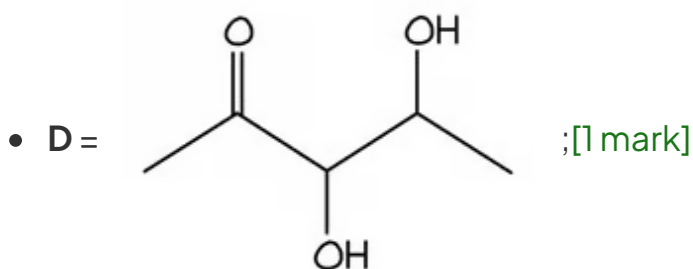
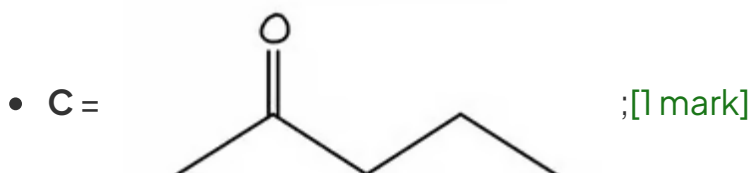
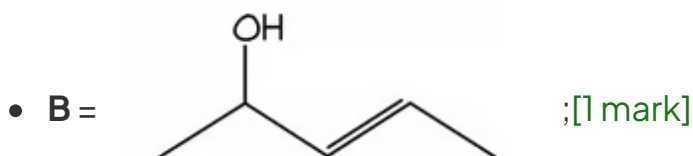
- Reaction **3** is an elimination reaction
 - Depending on your answer for compound **V**, 1-bromobutane or 2-bromobutane are reacting to form but-1-ene ($\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{Br}$ will form $\text{CH}_3\text{CH}_2\text{CH}=\text{CH}_2$) so a H and Br atom must be eliminated from the molecule
- This reaction requires extreme conditions so heat is required in alcoholic

conditions

- In aqueous conditions, nucleophilic substitution will occur

3a

a) The correct skeletal structures for compounds **B**, **C** and **D** are:



[Total: 3 marks]

- Compound **A** contains a C=O group (ketone) and a carbon=carbon double bond (alkene)
- The reaction of LiAlH_4 with compound **A** will reduce the ketone group (C=O) to an OH group but will not react with the C=C group
- H_2 and Ni will react with the C=C double bond and saturate it
- Cold, acidified KMnO_4 (aq) will react with alkenes to form a diol

3b

b) The colour change is:

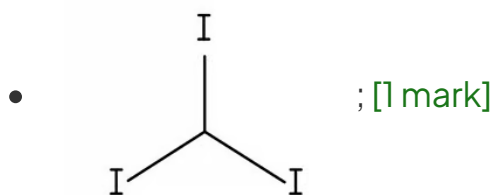
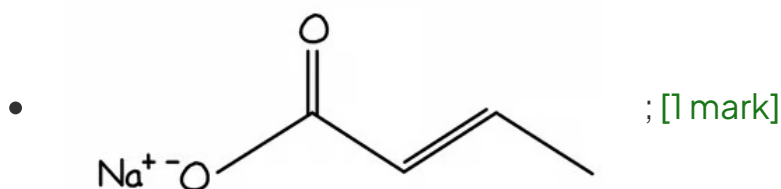
- Purple to colourless; [1 mark]

[Total: 1 mark]

- For the reactions of alkenes with cold, acidified potassium manganate, the colour change is purple to colourless
- The same colour change occurs during the oxidation of alcohols to aldehydes, carboxylic acids and ketones

3c

c) The skeletal structures of both these compounds are:



[Total: 2 marks]

- This reaction is called the iodoform reaction
- It forms a salt and tri-iodomethane
- The question states that iodine and hydroxide ions react with compound **A** to form two products
 - Tri-iodomethane has the formula CHI_3 and is a yellow precipitate
- The overall reaction is:



3b

b) The colour change is:

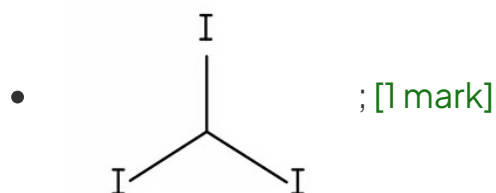
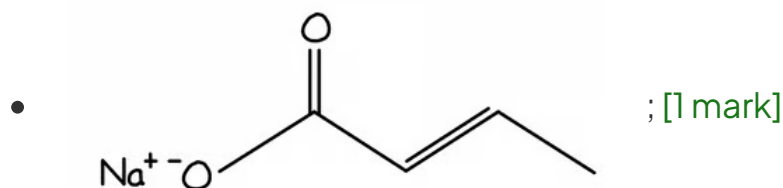
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[Total: 1 mark]

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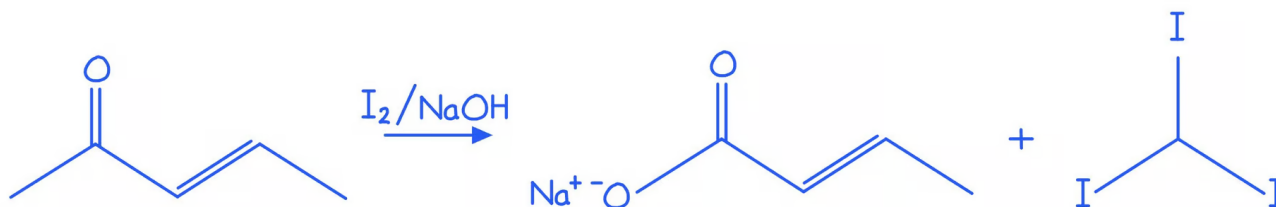
3c

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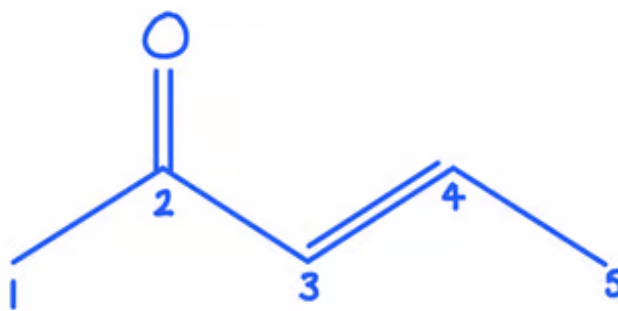
3d

d) The name of compound **A** is

- Pent-3-en(e)-2-one
- OR**
- 3-penten-2-one; [1 mark]

[Total: 1 mark]

- In compound **A**, there are 5 carbons in the main chain, therefore you must use the prefix 'pent'
- The carbonyl group is the highest priority group on the second carbon so '2-one' will be the suffix
- Compound **A** contains a $C=C$ on the third carbon atom, so '3-ene' should be in the name



4a

a) The mechanisms for each step are:

- Reaction **1** = electrophilic addition; [1 mark]
- Reaction **3** = nucleophilic addition; [1 mark]

[Total: 2 marks]

- Propene to propanol involves the electrophilic addition of alkenes
- Compound **G** has to be a ketone to form the hydroxynitrile, therefore the mechanism is nucleophilic addition

4b

b) The reagents and conditions are:

Reaction 1

- Steam; [1 mark]
- Phosphoric acid; [1 mark]

Reaction 2

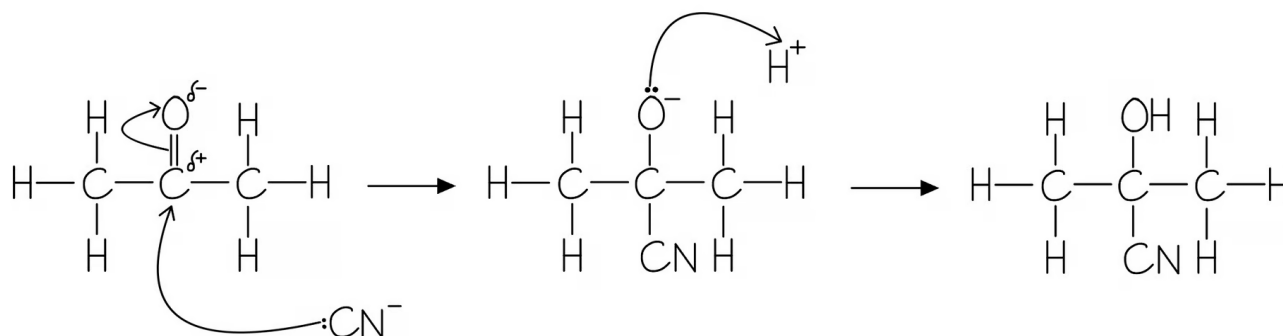
- Acidified potassium dichromate
OR
Acidified potassium manganate; [1 mark]
- Heat
AND
Reflux; [1 mark]

[Total: 4 marks]

- You could also be asked the colour changes for the oxidation of an alcohol
- For acidified potassium dichromate it is orange to green and for acidified potassium manganate it is purple to colourless
- Propan-2-ol is a secondary alcohol so will oxidise to form a ketone
- Heating under reflux is required for this reaction

4c

The reaction mechanism is:



- Correct partial charges, charges and lone pairs on atoms; [1 mark]
- Curly arrow from lone pair on CN nucleophile to δ^+ C atom in C=O bond; [1 mark]
- Curly arrow from C=O bond to O atom; [1 mark]
- Curly arrow from lone pair on O atom to H^+ ion; [1 mark]

[Total: 4 marks]

- The nucleophilic addition of hydrogen cyanide to carbonyl compounds is a two-step process
 - HCN is formed by the reaction of KCN or NaCN with sulfuric acid
 - It is toxic so is formed 'in situ'
- In step 1, the cyanide ion attacks the δ^+ carbonyl carbon to form a negatively charged intermediate
 - You must show the negative charge on the intermediate
- In step 2, the negatively charged oxygen atom in the reactive intermediate quickly reacts with aqueous H^+ (either from HCN, water or dilute acid) to form a 2-hydroxynitrile

4d

d) 2-hydroxy-2-methylpropanenitrile does not exhibit optical isomerism because:

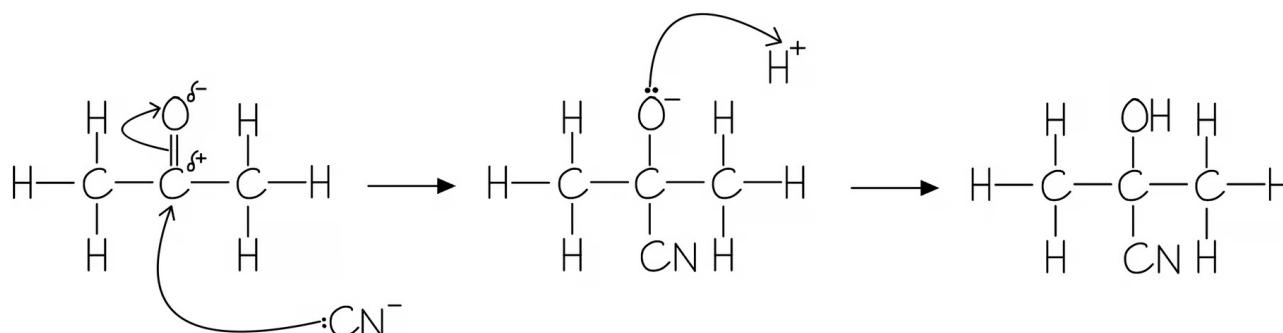
- No chiral carbon; [1 mark]
- As 2 of the groups / 2 methyl groups on carbon atom; [1 mark]

[Total: 2 marks]

- Propan-2-ol is a secondary alcohol so will oxidise to form a ketone
- Heating under reflux is required for this reaction

4c

The reaction mechanism is:



- Correct partial charges, charges and lone pairs on atoms; [1 mark]
- Curly arrow from lone pair on CN nucleophile to δ⁺C atom in C=O bond; [1 mark]
- Curly arrow from C=O bond to O atom; [1 mark]
- Curly arrow from lone pair on O atom to H⁺ ion; [1 mark]

[Total: 4 marks]

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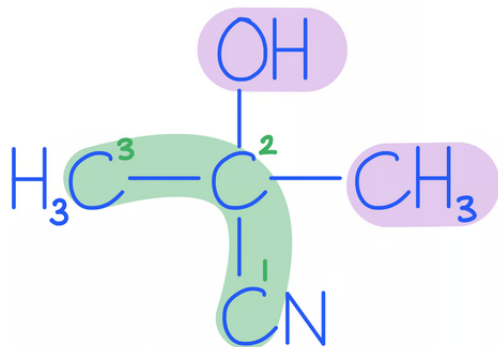
4d

d) 2-hydroxy-2-methylpropanenitrile does not exhibit optical isomerism because:

- No chiral carbon; [1 mark]
- As 2 of the groups / 2 methyl groups on carbon atom; [1 mark]

[Total: 2 marks]

- Optical isomers must contain a chiral carbon
- A chiral carbon has 4 different atoms or groups of atoms bonded to it
 - In 2-hydroxy-2-methylpropanenitrile, there are two methyl groups
- It can help to draw out the structure and identify the main carbon chain
 - In nitriles, the CN carbon is the first carbon



5a

a) Suitable reagents for the synthesis of Compound **A** via step **1** are:

- K₂Cr₂O₇ / Cr₂O₇²⁻ / (potassium) dichromate
OR
KMnO₄ / MnO₄⁻ / (potassium) permanganate; [1 mark]
- H₂SO₄ / H⁺ / acidified / with acid
OR
Heat; [1 mark]
- Oxidation; [1 mark]

[Total: 3 marks]

- An ester is made from a carboxylic acid and alcohol
- A carboxylic acid cannot be produced from the second molecule which is a ketone, so the first step in producing **A** must be the oxidation of an aldehyde to carboxylic acid

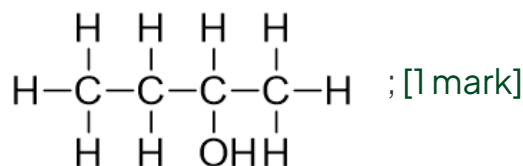
5b

b)

i) The name and structure of the molecule that is produced from step **2** are:

ii) The name and structure of the molecule that is produced from step 2 are.

- Butan-2-ol; [1 mark]
-



iii) The name of the type of reaction that is involved in step 2 and a suitable reagent for the process are

- Reduction; [1 mark]
- LiAlH_4 / lithium aluminium hydride / lithium tetrahydridoaluminate

OR

NaBH_4 / sodium borohydride / sodium tetrahydridoborate; [1 mark]

[Total: 4 marks]

- The reduction of a ketone produces a secondary alcohol
 - This secondary alcohol is required to produce the branched chain ester
- Lithium tetrahydridoaluminate, LiAlH_4 , is much more reactive than sodium tetrahydridoborate, NaBH_4 , so is more dangerous to use in the laboratory
- For the purposes of this question, both reagents are suitable answers

5c

c) The synthesis of ethanol from ethane in two steps:

Step 1: Ethane to chloroethane

- Reagent: chlorine / Cl_2 ; [1 mark]
- Conditions: UV light / high temps; [1 mark]
- Reaction type: (free radical) substitution / halogenation; [1 mark]

Step 2: Chloroethane to ethanol

- Reagent: aqueous sodium hydroxide / NaOH (aq) / aqueous potassium hydroxide / KOH (aq) ; [1 mark]
- Condition hot / heat under reflux; [1 mark]
- Reaction type: (nucleophilic) substitution / hydrolysis; [1 mark]

[Total: 6 marks]

- The conversion of an alkane to a halogenoalkane would be very impractical industrially as it is hard to control the extent of halogenation and to prevent multiple products from forming, however, here it is included to test your ability to apply your knowledge of organic reactions to solve problems in synthesis
- The type of mechanisms are shown in brackets, but they are not essential for the mark

5d

d)

i) The names of four possible substances **A** to **D**:

- A = propanone, B = propan-2-ol, C = propene D = 1-bromopropane / 1-chloropropane / 1-iodopropane; [1 mark]

ii) The reagents and conditions for step **4** are:

- Aqueous sodium hydroxide / NaOH (aq)
OR
Aqueous potassium hydroxide / KOH (aq); [1 mark]
- Heat under reflux; [1 mark]

[Total: 3 marks]

- **B** must be secondary alcohol formed by reduction of a three carbon ketone, which gives **A** as propanone and **B** as propan-2-ol
- **C** can only be a three carbon alkene, so must be propene
- In order to finish with propan-1-ol, the halogen must be on the first carbon in **D**, so it will be a 1-*halo*propane

